

Lecture 2 — Regression Analysis

The Ordinary Least Squares Estimator (SW Chapter 4)

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Outline

- 1 The population regression line
- 2 Simple regression: the least squares problem
- 3 Application: California test scores
- 4 Measures of fit
- 5 Multiple regression
- 6 Functional form
- 7 Decomposing means and variances

Where regression comes from

- So far we compared two conditional means: average test scores in districts with small vs. large classes. We found a gap of 7.4 points, statistically significant.
- But the policy question was: what is the effect of reducing the STR by *one* student per class? A two-bin comparison cannot answer that.
- **Linear regression** lets us estimate the whole *population regression line* and its slope — the difference in the expected value of Y for two values of X that differ by one unit.
- The estimated regression can be used for
 - **causal inference** (the causal effect on Y of a change in X), or
 - **prediction** (predicting Y given X for an observation not in the data).
- These place different requirements on the data — but both use the same toolkit.

The same three questions as before

The problem of statistical inference for linear regression is, at a general level, the same as for the mean or the difference between two means:

- **Estimation.** How should we draw a line through the data to estimate the population slope? *Answer: ordinary least squares (OLS).* What are its advantages and disadvantages?
- **Hypothesis testing.** How do we test whether the slope is zero?
- **Confidence intervals.** How do we construct a confidence interval for the slope?

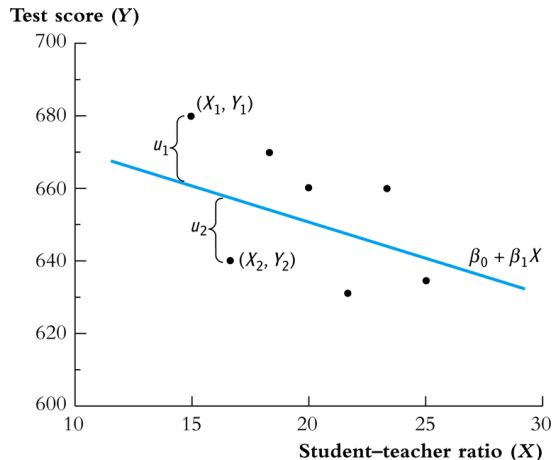
This lecture answers the first question — the *algebra* of OLS. The next lecture treats OLS as a random variable and answers the other two.

The population linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n$$

- X is the *independent variable* or *regressor*; Y is the *dependent variable*.
- β_0 is the intercept; β_1 is the slope of the population regression line.
- u_i is the **regression error**: it consists of omitted factors — other things that influence Y besides X — plus any error in measuring Y .
- Our running example: $TestScore = \beta_0 + \beta_1 STR + u$.
- β_0 and β_1 are *population* parameters. We would like to know β_1 ; we don't, so we must estimate it from data.

The model in a picture



- Seven observations (X_i, Y_i) , the population regression line $\beta_0 + \beta_1 X$, and the errors.
- u_1 is the vertical distance from the *line* to the point (X_1, Y_1) — not from the fitted line, which we have not computed yet.
- The line is a statement about the population; the points are one sample from it.
- Everything in this lecture is about picking a good line when you can see only the points.

Source: Stock & Watson, *Introduction to Econometrics*, 4th ed. (Global Edition) — Figure 4.1

The problem

Given a sample $\{(y_i, x_i) \in \mathbb{R}, i = 1, \dots, N\}$:

Find $\hat{a}, \hat{b} \in \mathbb{R}$ making $\hat{y}_i = \hat{a} + \hat{b}x_i$ as close as possible to y_i for all i .

In general one cannot find a, b solving

$$y_1 = a + bx_1, \quad y_2 = a + bx_2, \quad \dots, \quad y_N = a + bx_N$$

for more than two equations. How can we proceed?

Answer: give up on solving them exactly, and instead minimize the total size of the misses — measured by their squares.

Least squares

The least squares problem seeks $\hat{a}, \hat{b}$ minimizing the sum of squared residuals

$$\text{SSR}(a, b) = \sum_{i=1}^N (y_i - a - bx_i)^2, \quad (\hat{a}, \hat{b}) = \arg \min_{a, b} \text{SSR}(a, b).$$

The first-order conditions are

$$\begin{aligned} \frac{\partial \text{SSR}(\hat{a}, \hat{b})}{\partial a} &= -2 \sum_{i=1}^N (y_i - \hat{a} - \hat{b}x_i) = 0, \\ \frac{\partial \text{SSR}(\hat{a}, \hat{b})}{\partial b} &= -2 \sum_{i=1}^N x_i (y_i - \hat{a} - \hat{b}x_i) = 0. \end{aligned}$$

- (Make sure you understand that the derivative of $b \mapsto (y_i - a - bx_i)^2$ is $-2x_i(y_i - a - bx_i)$.)
- $\text{SSR}(a, b)$ is convex, so the FOCs are sufficient: the minimum is unique and global. These FOCs are the **normal equations**.

Normal equations = moment conditions

Writing $\bar{x}, \bar{y}, \overline{x^2}, \overline{xy}$ for the sample means, the FOCs become

$$\begin{cases} \bar{y} = \hat{a} + \hat{b} \bar{x}, \\ \overline{xy} = \hat{a} \bar{x} + \hat{b} \overline{x^2}. \end{cases}$$

- They are called **normal equations** because they arise from minimizing the sum of squared residuals.
- Equivalently they are **moment conditions**: with $\hat{u}_i = y_i - \hat{a} - \hat{b}x_i$,

$$\sum_{i=1}^N \hat{u}_i = 0 \quad \text{and} \quad \sum_{i=1}^N x_i \hat{u}_i = 0.$$

- The population counterpart is $\mathbb{E}[Xu] = 0$. If $\mathbb{E}[u] = 0$ then $\text{Cov}(X, u) = \mathbb{E}[Xu] - \mathbb{E}[X]\mathbb{E}[u] = \mathbb{E}[Xu]$, so the moment condition says the error is **uncorrelated with (orthogonal to) the regressor**. Remember this — it is the assumption everything in the next lecture rests on.

The OLS estimator

Multiply the first normal equation by $\bar{x}$ and subtract from the second:

$$\overline{xy} - \bar{x} \bar{y} = \hat{b}(\overline{x^2} - \bar{x}^2) \iff \hat{b} = \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2},$$

$$\hat{a} = \bar{y} - \hat{b} \bar{x}.$$

- The regression line has slope $\hat{b}$ and passes through the centroid $(\bar{x}, \bar{y})$ of the scatterplot.
- In SW's notation: $\hat{\beta}_1 = s_{XY}/s_X^2$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$.
- Note the parallel with Lecture 1: $\bar{Y}$ is the least squares estimator of μ_Y (it solves $\min_m \sum (Y_i - m)^2$); OLS is the same idea with a line instead of a constant.

Interpretation of the regression coefficient

For a line $y = a + bx$, $b = dy/dx$ is the change in y from a one-unit change in x .

Think of regressing log wages on years of education: $\hat{b}$ measures the change in $\ln w$ from one additional year of education.

- If $y \approx y'$, then $\ln y' - \ln y \approx (y' - y)/y$ is the relative wage growth.
- Regressing log consumption on log income gives the elasticity of consumption to income.
- *Rule of thumb*: always log positive continuous variables. Do not log discrete counting variables (like whole years of education) — a one-unit change is not infinitesimal for a count.

Two particular cases

Model with no regressor:

$$\min_a \sum_{i=1}^N (y_i - a)^2 \Rightarrow \hat{a} = \bar{y}.$$

Model with no intercept:

$$\min_b \sum_{i=1}^N (y_i - bx_i)^2 \Rightarrow \hat{b} = \frac{\sum_i x_i y_i}{\sum_i x_i^2} = \frac{\overline{xy}}{\overline{x^2}}.$$

The first case is why the sample mean is “the regression of Y on a constant” — and why everything you learned about $\bar{Y}$ carries over.

Centering and standardization

Centering. Replace y_i, x_i by $y_i - \bar{y}$ and $x_i - \bar{x}$. Regressing the centered variables *without* an intercept gives the same slope, $\hat{b} = \text{Cov}(X, Y) / \text{Var}(X)$.

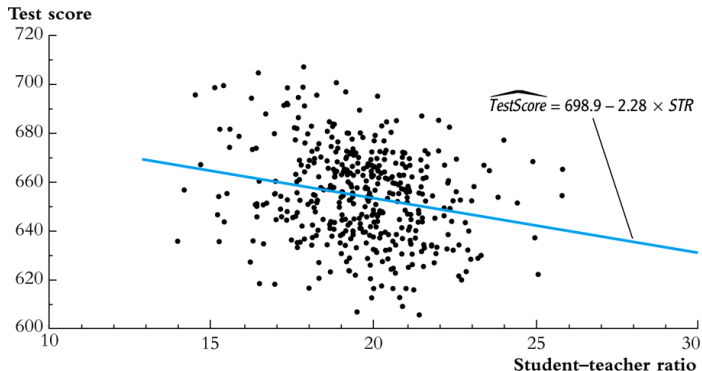
- Adding a constant is equivalent to centering all variables first. Unless all variables are centered, **always include an intercept.**

Standardization. Replace y_i, x_i by $y_i / \text{sd}(Y)$ and $x_i / \text{sd}(X)$. Then

$$\hat{b} = \frac{\text{Cov}(X, Y)}{\text{sd}(X) \text{sd}(Y)} = \text{Corr}(X, Y).$$

- It is common to standardize X but not Y . Then the coefficient becomes $\hat{b} \text{sd}(X)$: the effect on Y of a *one-standard-deviation* change in X . Choose a meaningful unit for X .

Application to the California class size data



- The OLS line through the 420 districts: $\widehat{TestScore} = 698.9 - 2.28 \times STR$.
- This is the same scatterplot we have seen since Lecture 1 — now with the least squares line drawn through it.

Source: Stock & Watson, *Introduction to Econometrics*, 4th ed. (Global Edition) — Figure 4.2

Interpreting the estimated slope and intercept

$$\widehat{TestScore} = 698.9 - 2.28 \times STR$$

- **Slope.** Districts with one more student per teacher on average have test scores that are 2.28 points lower.
- Equivalently: a reduction of 2 students per class is associated with about 4.6 points — roughly a quarter of a standard deviation of test scores (19.1).
- **Intercept.** Taken literally, districts with *zero* students per teacher would have a predicted test score of 698.9. This interpretation makes no sense — it extrapolates the line far outside the range of the data. Here the intercept is not economically meaningful.
- An intercept is almost always necessary for the *fit*, and almost never interesting in *itself*, unless $X = 0$ is inside the data.

Predicted values and residuals: one district

- One of the districts is **Antelope, CA**, with $STR = 19.33$ and $TestScore = 657.8$.
- Its predicted value is

$$\widehat{TestScore}_{Antelope} = 698.9 - 2.28 \times 19.33 = 654.8,$$

- and its residual is

$$\hat{u}_{Antelope} = 657.8 - 654.8 = 3.0.$$

- Antelope did 3 points better than the regression line predicts for a district with its class size. *Why?* Whatever the reason — teacher quality, family income, measurement noise — it lives in u . That is exactly the object the next lecture makes assumptions about.

OLS regression: STATA output

```
. regress testscr str, robust
```

```
Regression with robust standard errors Number of obs = 420
                                F( 1, 418) = 19.26
                                Prob > F = 0.0000
                                R-squared = 0.0512
                                Root MSE = 18.581
```

```
-----+-----
      | Robust
testscr| Coef. Std. Err. t P>|t| [95% Conf. Interval]
-----+-----
      str | -2.279808 .5194892 -4.39 0.000 -3.300945 -1.258671
      _cons | 698.933 10.36436 67.44 0.000 678.5602 719.3057
-----+-----
```

- Coef. gives $\hat{\beta}_1 = -2.28$ and $\hat{\beta}_0 = 698.9$.
- Every other column — standard error, t , p -value, confidence interval — belongs to the *next* lecture. Ignore them for now.

Predictions and residuals: the algebra

Define $\hat{y}_i = \hat{a} + \hat{b}x_i$ and $\hat{u}_i = y_i - \hat{y}_i$.

- **Mean residual is zero** (first normal equation): $\frac{1}{N} \sum_i \hat{u}_i = 0$.
- Hence **mean prediction equals mean outcome**:

$$\frac{1}{N} \sum_i \hat{y}_i = \hat{a} + \hat{b} \bar{x} = \bar{y}.$$

- **Prediction and residual are uncorrelated**:

$$\text{Cov}(\hat{Y}, \hat{U}) = \frac{1}{N} \sum_i (\hat{y}_i - \bar{y}) \hat{u}_i = \frac{1}{N} \sum_i \hat{y}_i \hat{u}_i - \bar{y} \cdot \frac{1}{N} \sum_i \hat{u}_i = 0,$$

$\underbrace{\qquad\qquad\qquad}_{=0} \qquad\qquad\qquad \underbrace{\qquad\qquad\qquad}_{=0}$

because $\frac{1}{N} \sum_i \hat{y}_i \hat{u}_i = \hat{a} \frac{1}{N} \sum_i \hat{u}_i + \hat{b} \frac{1}{N} \sum_i x_i \hat{u}_i = 0$ by the two normal equations.

- These are *algebraic* facts. They hold in every sample, whatever the model is.

Analysis of variance

By construction $Y = \hat{Y} + \hat{U}$, and $\text{Cov}(\hat{Y}, \hat{U}) = 0$, so

$$\text{Var}(Y) = \text{Var}(\hat{Y}) + \text{Var}(\hat{U}).$$

In sums of squares (SW's notation):

- $\text{TSS} = \sum_i (y_i - \bar{y})^2$ *(total sum of squares)*
- $\text{ESS} = \sum_i (\hat{y}_i - \bar{y})^2$ *(explained sum of squares)*
- $\text{SSR} = \sum_i \hat{u}_i^2$ *(sum of squared residuals)*

$$\text{TSS} = \text{ESS} + \text{SSR}$$

- **Warning on notation.** Different books swap these letters (you will see SSE used for both “explained” and “error”). We follow SW throughout: **E**xplained, **R**esidual, **T**otal.

The regression R^2

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{SSR}}{\text{TSS}} \in [0, 1]$$

- R^2 is the fraction of the sample variance of Y_i explained by the regression.
- $R^2 = 0$ means $\text{ESS} = 0$; $R^2 = 1$ means $\text{ESS} = \text{TSS}$.
- For a regression with a single X , R^2 is exactly the *square of the correlation coefficient* between X and Y .
- It is unitless. A good fit requires a large R^2 ; it tends to be larger for time series than for cross-sections. A small R^2 does not mean the model is useless.

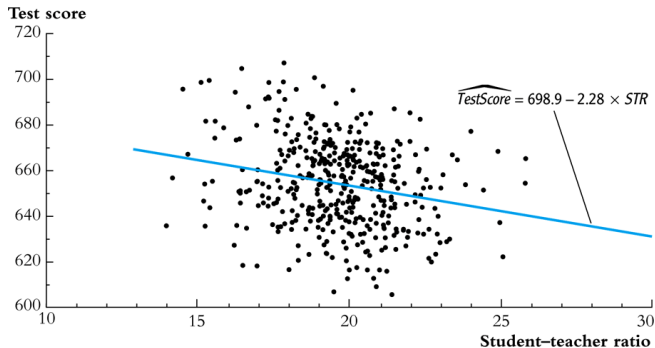
The standard error of the regression (SER)

The SER measures the spread of the distribution of u — it is (almost) the sample standard deviation of the OLS residuals:

$$SER = s_{\hat{u}} = \sqrt{\frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2} = \sqrt{\frac{SSR}{n-2}}$$

- It has the **units of Y** : it measures the average “size” of the OLS residual — the average mistake made by the regression line.
- The root mean squared error (RMSE) divides by n instead of $n-2$; it measures the same thing.
- *Why $n-2$?* Two coefficients have been estimated, so two degrees of freedom are used up (same logic as the $n-1$ in s_Y^2). We prove $\mathbb{E}[SSR] = (n-2)\sigma^2$ next lecture. When n is large it does not matter whether you use n , $n-1$ or $n-2$.

R^2 and SER in the class size example



$$R^2 = 0.051, \quad SER = 18.6$$

- STR explains only about 5% of the variation in test scores. The typical district is 18.6 points off the line.
- Does this make sense? Look at the scatter: yes — districts at the same STR differ enormously.
- **Does this mean the STR is unimportant in a policy sense?** No. R^2 answers “how well can I predict Y ?”; policy asks “what happens if I change X ?” Those are different questions, and a small R^2 is not evidence against a real effect.

Definition

We want to learn about the link between Y and variables $X_1, \dots, X_K$. The simplest relationship is linear:

$$Y = a + b_1X_1 + b_2X_2 + \dots + b_KX_K.$$

This is the regression of the dependent variable Y on regressors $X_1, \dots, X_K$ with an intercept.

Example: the Mincer wage equation (data: U.S. CPS). $Y = \log$ income; the regressors capture human capital — X_1 years of education, X_2 potential experience, X_3 its square, and so on.

Why we need it. In the class-size regression, u contains family income, English learners, teacher quality. . . all plausibly correlated with STR. Multiple regression is how we take some of them out of u and put them on the right-hand side.

OLS with several regressors

Sample $\{(y_i, x_{1i}, \dots, x_{Ki})\}_{i=1}^N$; prediction and residual

$$\hat{y}_i = \hat{a} + \hat{b}_1 x_{1i} + \dots + \hat{b}_K x_{Ki}, \quad \hat{u}_i = y_i - \hat{y}_i.$$

OLS minimizes $\sum_i \hat{u}_i^2$. The normal equations are

$$\frac{1}{N} \sum_i \hat{u}_i = 0, \quad \frac{1}{N} \sum_i x_{ki} \hat{u}_i = 0 \quad (k = 1, \dots, K),$$

giving $K + 1$ linear equations for $K + 1$ unknowns — exactly the same moment conditions as before, one per regressor.

All the algebra from the simple case carries over unchanged: mean residual zero, mean prediction = $\bar{y}$, $\text{Cov}(\hat{Y}, \hat{U}) = 0$, $\text{TSS} = \text{ESS} + \text{SSR}$, $R^2 = \text{ESS}/\text{TSS}$.

Vector form

Write the model as $y = X\beta + u$, where y is $n \times 1$, X is $n \times k$ (including a column of ones), β is $k \times 1$ and u is $n \times 1$. OLS solves

$$\hat{\beta} = \arg \min_{\beta} (y - X\beta)'(y - X\beta).$$

The first-order condition is $-2X'(y - X\hat{\beta}) = 0$, i.e. the **normal equations**

$$X'X\hat{\beta} = X'y \quad \implies \quad \boxed{\hat{\beta} = (X'X)^{-1}X'y}$$

provided $X'X$ is invertible.

In population terms the corresponding moment condition is $\mathbb{E}[X'u] = 0$: each regressor is uncorrelated with the error term.

Suppose $x_{1i} + x_{2i} = 1$ for every i . If $(\hat{a}, \hat{b}_1, \hat{b}_2)$ solves LS, then

$$\hat{a} + \hat{b}_1 x_{1i} + \hat{b}_2 x_{2i} = \hat{a}(x_{1i} + x_{2i}) + \hat{b}_1 x_{1i} + \hat{b}_2 x_{2i} = (\hat{b}_1 + \hat{a})x_{1i} + (\hat{b}_2 + \hat{a})x_{2i},$$

so $(0, \hat{b}_1 + \hat{a}, \hat{b}_2 + \hat{a})$ is another solution. There is an infinity of LS minimizers $\Rightarrow$ the parameters are **not identified**.

Rank condition. The variables $(1, x_{1i}, \dots, x_{Ki})$ must be linearly independent: there is no $(\alpha, \beta_1, \dots, \beta_K) \neq 0$ with $\alpha + \beta_1 x_{1i} + \dots + \beta_K x_{Ki} = 0$ for all i . This requires $N \geq K + 1$.

In matrix terms: $\text{rank}(X) = k$, which is exactly what makes $X'X$ invertible.

Marginal effects

Interpret $\hat{b}_k$ as the marginal effect of x_k — the effect on y of a one-unit change in x_k , *holding the other x 's constant*:

$$b_k = \left. \frac{\Delta y}{\Delta x_k} \right|_{\text{others const.}} = \frac{\partial y}{\partial x_k}.$$

“Holding the others constant” is doing a lot of work in that sentence. It is what distinguishes a multiple-regression coefficient from a simple-regression one, and it is the whole reason we add controls.

Model a non-linear relationship between y and a continuous x (e.g. wage and age) with a polynomial:

$$y = a + b_1x + b_2x^2 + b_3x^3 + \dots, \quad \frac{dy}{dx} = b_1 + 2b_2x + 3b_3x^2 + \dots.$$

- A quadratic often suffices, and lets the marginal effect vary with x .
- For wage–age we expect $b_1 > 0$, $b_2 < 0$, with a peak at $x^* = -b_1/(2b_2)$.
- Note the model is still *linear in the parameters* — which is all OLS requires. “Linear regression” never meant a straight line in x .

Interaction effects

Suppose the effect of x_2 depends on x_1 : $\frac{\partial y}{\partial x_2} = b_2 + b_3 x_1$. E.g. does education raise the return to experience? This implies

$$y = a + b_1 x_1 + (b_2 + b_3 x_1) x_2 + \dots = a + b_1 x_1 + b_2 x_2 + b_3 x_1 x_2 + \dots$$

Symmetrically, $\frac{\partial y}{\partial x_1} = b_1 + b_3 x_2$: a positive b_3 can be read as higher returns to education for high-ability individuals.

A collinearity trap. In a cross-section, age and date of birth are collinear with calendar time: age = date – dob, so you cannot include all three linearly. Even in panels the three are collinear; non-linear transforms may help but are somewhat arbitrary.

Log transformation

Logarithms make estimation invariant to scale (unit) changes, since

$$a + b \log(\alpha x) = a + b \log \alpha + b \log x.$$

If the dependent variable is also log-transformed,

$$\log y = a + b \log x \Rightarrow b = \frac{d \log y}{d \log x} = \frac{dy/y}{dx/x},$$

the **elasticity** of y with respect to x (percent change in y from a 1% change in x).

Use logs for positive, continuous variables — not for discrete counts like whole years of education.

Discrete regressors: dummy coding

Suppose wage y and education $x \in \{1, 2, 3\}$ (1: less than high school; 2: some college; 3: college+).

- **Option 1:** treat x as quantitative (replace each category by mean years) $\Rightarrow$ loss of information, and it imposes equal spacing.
- **Option 2:** create dummies $x_1 = \mathbf{1}\{x = 1\}$, $x_2 = \mathbf{1}\{x = 2\}$, $x_3 = \mathbf{1}\{x = 3\}$.

Since $x_1 + x_2 + x_3 = 1$, you cannot regress y on x_1, x_2, x_3 *and* a constant — that is exactly the multicollinearity example above (the “dummy variable trap”). Remove one dummy: its category becomes the **reference** category.

Interpreting dummy coefficients

Drop x_3 and estimate $y(x) = a + b_1x_1 + b_2x_2$. The constant absorbs the effect of $x = 3$, so coefficients are differences relative to the reference:

$$y(1) = a + b_1, \quad y(2) = a + b_2, \quad y(3) = a \quad \implies \quad b_1 = y(1) - y(3), \quad b_2 = y(2) - y(3).$$

Typically $b_1 < b_2 < 0$.

Several discrete regressors. Let z be gender ($z = 1$ female, $z = 2$ male); define $z_1 = \mathbf{1}\{z = 1\}$ and drop z_2 . Run $y(x, z) = a + b_1x_1 + b_2x_2 + cz_1$. Then $y(3, 2) = a$ is the baseline (college-educated males) and $c = y(x, 1) - y(x, 2)$ is the gender gap, the same at every education level.

To let it vary, interact: $y = a + b_1x_1 + b_2x_2 + cz_1 + d_{11}(x_1z_1) + d_{21}(x_2z_1)$. Beware — the parameter count explodes quickly.

Decomposition of the mean

Each unit satisfies $y_i = \hat{a} + \hat{b}_1 x_{1i} + \cdots + \hat{b}_K x_{Ki} + \hat{u}_i$. Taking sample means (and using $\bar{\hat{u}} = 0$),

$$\bar{y} = \hat{a} + \hat{b}_1 \bar{x}_1 + \cdots + \hat{b}_K \bar{x}_K.$$

Example. If y_i is GDP growth, x_{1i} R&D and x_{2i} capital investment, then $\hat{b}_1 \bar{x}_1$ is innovation's contribution to average growth and $\hat{b}_2 \bar{x}_2$ the contribution of capital.

This is *growth accounting*: a regression turned into a table of contributions.

	Labour input	ICT capital	Non-ICT capital	Multi-factor prod.	GDP growth	
Italy	0.32	0.25	0.55	0.60	1.72	1985–2004
	0.49	0.18	0.54	-0.52	0.70	2001–2004
Germany	-0.33	0.28	0.30	1.17	1.42	1985–2004
	-0.35	0.21	0.13	0.89	0.88	2001–2004
Japan	-0.43	0.39	0.53	1.61	2.08	1985–2004
	-0.71	0.37	0.01	1.77	1.44	2001–2004
France	0.13	0.31	0.38	1.33	2.15	1985–2004
	0.08	0.26	0.31	0.97	1.63	2001–2004
UK	0.43	0.58	0.43	1.14	2.57	1985–2004
	0.27	0.49	0.31	1.36	2.42	2001–2004
US	1.03	0.49	0.36	1.09	2.94	1985–2004
	0.45	0.34	0.24	1.65	2.67	2001–2004
Canada	1.16	0.41	0.67	0.51	2.72	1985–2004
	1.29	0.33	0.60	0.47	2.68	2001–2004

Decomposition of the variance

Using $\text{Var}(X_1 + X_2) = \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2)$, $\text{Var}(a + bX) = b^2\text{Var}(X)$ and $\text{Cov}(b_1X_1, b_2X_2) = b_1b_2\text{Cov}(X_1, X_2)$,

$$\text{Var}(y_i) = \hat{b}_1^2 \text{Var}(x_{1i}) + \hat{b}_2^2 \text{Var}(x_{2i}) + 2\hat{b}_1\hat{b}_2\text{Cov}(x_{1i}, x_{2i}) + \cdots + \text{Var}(\hat{u}_i).$$

This is how one attributes inequality in y to inequality in each x — e.g. how much of the variance of log wages is accounted for by the variance of education.

Where we are, and what is missing

What we have done. Pure algebra. Given numbers (y_i, x_i) , OLS draws a line; R^2 and the SER describe how close it is. None of this required a model, a population, or any assumption at all.

What we have not done.

- Is $\hat{b}$ close to the true β_1 ? Is it even estimating anything?
- How much would $\hat{b}$ change in a different sample of 420 districts?
- Is -2.28 distinguishable from 0?
- Can we call -2.28 a *causal effect*?

Next lecture: treat $\hat{b}$ as a random variable — the least squares assumptions, unbiasedness, the variance of $\hat{\beta}_1$, the Gauss–Markov theorem, and t - and F -tests.